

PERMUTE-WITH-ALL(A, n)

```

1  for  $i = 1$  to  $n$ 
2      swap  $A[i]$  with  $A[\text{RANDOM}(1, n)]$ 

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5.3-4

Professor Knieval suggests the procedure PERMUTE-BY-CYCLE to generate a uniform random permutation. Show that each element $A[i]$ has a $1/n$ probability of winding up in any particular position in B . Then show that Professor Knieval is mistaken by showing that the resulting permutation is not uniformly random.

PERMUTE-BY-CYCLE(A, n)

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1  let  $B[1:n]$  be a new array
2   $offset = \text{RANDOM}(1, n)$ 
3  for  $i = 1$  to  $n$ 
4       $dest = i + offset$ 
5      if  $dest > n$ 
6           $dest = dest - n$ 
7       $B[dest] = A[i]$ 
8  return  $B$ 

```

5.3-5

Professor Gallup wants to create a *random sample* of the set $\{1, 2, 3, \dots, n\}$, that is, an m -element subset S , where $0 \leq m \leq n$, such that each m -subset is equally likely to be created. One way is to set $A[i] = i$, for $i = 1, 2, 3, \dots, n$, call RANDOMLY-PERMUTE(A), and then take just the first m array elements. This method makes n calls to the RANDOM procedure. In Professor Gallup's application, n is much larger than m , and so the professor wants to create a random sample with fewer calls to RANDOM.

RANDOM-SAMPLE(m, n)

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1   $S = \emptyset$ 
2  for  $k = n - m + 1$  to  $n$            // iterates  $m$  times
3       $i = \text{RANDOM}(1, k)$ 
4      if  $i \in S$ 
5           $S = S \cup \{k\}$ 
6      else  $S = S \cup \{i\}$ 
7  return  $S$ 

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Show that the procedure RANDOM-SAMPLE on the previous page returns a random m -subset S of $\{1, 2, 3, \dots, n\}$, in which each m -subset is equally likely, while making only m calls to RANDOM.

★ 5.4 Probabilistic analysis and further uses of indicator random variables

This advanced section further illustrates probabilistic analysis by way of four examples. The first determines the probability that in a room of k people, two of them share the same birthday. The second example examines what happens when randomly tossing balls into bins. The third investigates “streaks” of consecutive heads when flipping coins. The final example analyzes a variant of the hiring problem in which you have to make decisions without actually interviewing all the candidates.

5.4.1 The birthday paradox

Our first example is the *birthday paradox*. How many people must there be in a room before there is a 50% chance that two of them were born on the same day of the year? The answer is surprisingly few. The paradox is that it is in fact far fewer than the number of days in a year, or even half the number of days in a year, as we shall see.

To answer this question, we index the people in the room with the integers $1, 2, \dots, k$, where k is the number of people in the room. We ignore the issue of leap years and assume that all years have $n = 365$ days. For $i = 1, 2, \dots, k$, let b_i be the day of the year on which person i 's birthday falls, where $1 \leq b_i \leq n$. We also assume that birthdays are uniformly distributed across the n days of the year, so that $\Pr\{b_i = r\} = 1/n$ for $i = 1, 2, \dots, k$ and $r = 1, 2, \dots, n$.

The probability that two given people, say i and j , have matching birthdays depends on whether the random selection of birthdays is independent. We assume from now on that birthdays are independent, so that the probability that i 's birthday and j 's birthday both fall on day r is

$$\begin{aligned} \Pr\{b_i = r \text{ and } b_j = r\} &= \Pr\{b_i = r\} \Pr\{b_j = r\} \\ &= \frac{1}{n^2}. \end{aligned}$$

Thus, the probability that they both fall on the same day is

$$\Pr\{b_i = b_j\} = \sum_{r=1}^n \Pr\{b_i = r \text{ and } b_j = r\}$$

$$\begin{aligned}
&= \sum_{r=1}^n \frac{1}{n^2} \\
&= \frac{1}{n}.
\end{aligned} \tag{5.7}$$

More intuitively, once b_i is chosen, the probability that b_j is chosen to be the same day is $1/n$. As long as the birthdays are independent, the probability that i and j have the same birthday is the same as the probability that the birthday of one of them falls on a given day.

We can analyze the probability of at least 2 out of k people having matching birthdays by looking at the complementary event. The probability that at least two of the birthdays match is 1 minus the probability that all the birthdays are different. The event B_k that k people have distinct birthdays is

$$B_k = \bigcap_{i=1}^k A_i,$$

where A_i is the event that person i 's birthday is different from person j 's for all $j < i$. Since we can write $B_k = A_k \cap B_{k-1}$, we obtain from equation (C.18) on page 1189 the recurrence

$$\Pr\{B_k\} = \Pr\{B_{k-1}\} \Pr\{A_k \mid B_{k-1}\}, \tag{5.8}$$

where we take $\Pr\{B_1\} = \Pr\{A_1\} = 1$ as an initial condition. In other words, the probability that $b_1, b_2, \dots, b_k$ are distinct birthdays equals the probability that $b_1, b_2, \dots, b_{k-1}$ are distinct birthdays multiplied by the probability that $b_k \neq b_i$ for $i = 1, 2, \dots, k-1$, given that $b_1, b_2, \dots, b_{k-1}$ are distinct.

If $b_1, b_2, \dots, b_{k-1}$ are distinct, the conditional probability that $b_k \neq b_i$ for $i = 1, 2, \dots, k-1$ is $\Pr\{A_k \mid B_{k-1}\} = (n - k + 1)/n$, since out of the n days, $n - (k - 1)$ days are not taken. We iteratively apply the recurrence (5.8) to obtain

$$\begin{aligned}
\Pr\{B_k\} &= \Pr\{B_{k-1}\} \Pr\{A_k \mid B_{k-1}\} \\
&= \Pr\{B_{k-2}\} \Pr\{A_{k-1} \mid B_{k-2}\} \Pr\{A_k \mid B_{k-1}\} \\
&\vdots \\
&= \Pr\{B_1\} \Pr\{A_2 \mid B_1\} \Pr\{A_3 \mid B_2\} \cdots \Pr\{A_k \mid B_{k-1}\} \\
&= 1 \cdot \left(\frac{n-1}{n}\right) \left(\frac{n-2}{n}\right) \cdots \left(\frac{n-k+1}{n}\right) \\
&= 1 \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).
\end{aligned}$$

Inequality (3.14) on page 66, $1 + x \leq e^x$, gives us

$$\begin{aligned}
\Pr\{B_k\} &\leq e^{-1/n} e^{-2/n} \dots e^{-(k-1)/n} \\
&= e^{-\sum_{i=1}^{k-1} i/n} \\
&= e^{-k(k-1)/2n} \\
&\leq \frac{1}{2}
\end{aligned}$$

when $-k(k-1)/2n \leq \ln(1/2)$. The probability that all k birthdays are distinct is at most $1/2$ when $k(k-1) \geq 2n \ln 2$ or, solving the quadratic equation, when $k \geq (1 + \sqrt{1 + (8 \ln 2)n})/2$. For $n = 365$, we must have $k \geq 23$. Thus, if at least 23 people are in a room, the probability is at least $1/2$ that at least two people have the same birthday. Since a year on Mars is 669 Martian days long, it takes 31 Martians to get the same effect.

An analysis using indicator random variables

Indicator random variables afford a simpler but approximate analysis of the birthday paradox. For each pair (i, j) of the k people in the room, define the indicator random variable X_{ij} , for $1 \leq i < j \leq k$, by

$$\begin{aligned}
X_{ij} &= \mathbf{I}\{\text{person } i \text{ and person } j \text{ have the same birthday}\} \\
&= \begin{cases} 1 & \text{if person } i \text{ and person } j \text{ have the same birthday,} \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

By equation (5.7), the probability that two people have matching birthdays is $1/n$, and thus by Lemma 5.1 on page 130, we have

$$\begin{aligned}
\mathbb{E}[X_{ij}] &= \Pr\{\text{person } i \text{ and person } j \text{ have the same birthday}\} \\
&= 1/n.
\end{aligned}$$

Letting X be the random variable that counts the number of pairs of individuals having the same birthday, we have

$$X = \sum_{i=1}^{k-1} \sum_{j=i+1}^k X_{ij}.$$

Taking expectations of both sides and applying linearity of expectation, we obtain

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}\left[\sum_{i=1}^{k-1} \sum_{j=i+1}^k X_{ij}\right] \\
&= \sum_{i=1}^{k-1} \sum_{j=i+1}^k \mathbb{E}[X_{ij}]
\end{aligned}$$

$$\begin{aligned}
&= \binom{k}{2} \frac{1}{n} \\
&= \frac{k(k-1)}{2n}.
\end{aligned}$$

When $k(k-1) \geq 2n$, therefore, the expected number of pairs of people with the same birthday is at least 1. Thus, if we have at least $\sqrt{2n} + 1$ individuals in a room, we can expect at least two to have the same birthday. For $n = 365$, if $k = 28$, the expected number of pairs with the same birthday is $(28 \cdot 27)/(2 \cdot 365) \approx 1.0356$. Thus, with at least 28 people, we expect to find at least one matching pair of birthdays. On Mars, with 669 days per year, we need at least 38 Martians.

The first analysis, which used only probabilities, determined the number of people required for the probability to exceed $1/2$ that a matching pair of birthdays exists, and the second analysis, which used indicator random variables, determined the number such that the expected number of matching birthdays is 1. Although the exact numbers of people differ for the two situations, they are the same asymptotically: $\Theta(\sqrt{n})$.

5.4.2 Balls and bins

Consider a process in which you randomly toss identical balls into b bins, numbered $1, 2, \dots, b$. The tosses are independent, and on each toss the ball is equally likely to end up in any bin. The probability that a tossed ball lands in any given bin is $1/b$. If we view the ball-tossing process as a sequence of Bernoulli trials (see Appendix C.4), where success means that the ball falls in the given bin, then each trial has a probability $1/b$ of success. This model is particularly useful for analyzing hashing (see Chapter 11), and we can answer a variety of interesting questions about the ball-tossing process. (Problem C-2 asks additional questions about balls and bins.)

- *How many balls fall in a given bin?* The number of balls that fall in a given bin follows the binomial distribution $b(k; n, 1/b)$. If you toss n balls, equation (C.41) on page 1199 tells us that the expected number of balls that fall in the given bin is n/b .
- *How many balls must you toss, on the average, until a given bin contains a ball?* The number of tosses until the given bin receives a ball follows the geometric distribution with probability $1/b$ and, by equation (C.36) on page 1197, the expected number of tosses until success is $1/(1/b) = b$.
- *How many balls must you toss until every bin contains at least one ball?* Let us call a toss in which a ball falls into an empty bin a “hit.” We want to know the expected number n of tosses required to get b hits.

Using the hits, we can partition the n tosses into stages. The i th stage consists of the tosses after the $(i - 1)$ st hit up to and including the i th hit. The first stage consists of the first toss, since you are guaranteed to have a hit when all bins are empty. For each toss during the i th stage, $i - 1$ bins contain balls and $b - i + 1$ bins are empty. Thus, for each toss in the i th stage, the probability of obtaining a hit is $(b - i + 1)/b$.

Let n_i denote the number of tosses in the i th stage. The number of tosses required to get b hits is $n = \sum_{i=1}^b n_i$. Each random variable n_i has a geometric distribution with probability of success $(b - i + 1)/b$ and thus, by equation (C.36), we have

$$E[n_i] = \frac{b}{b - i + 1}.$$

By linearity of expectation, we have

$$\begin{aligned} E[n] &= E\left[\sum_{i=1}^b n_i\right] \\ &= \sum_{i=1}^b E[n_i] \\ &= \sum_{i=1}^b \frac{b}{b - i + 1} \\ &= b \sum_{i=1}^b \frac{1}{i} \quad (\text{by equation (A.14) on page 1144}) \\ &= b(\ln b + O(1)) \quad (\text{by equation (A.9) on page 1142}). \end{aligned}$$

It therefore takes approximately $b \ln b$ tosses before we can expect that every bin has a ball. This problem is also known as the *coupon collector's problem*, which says that if you are trying to collect each of b different coupons, then you should expect to acquire approximately $b \ln b$ randomly obtained coupons in order to succeed.

5.4.3 Streaks

Suppose that you flip a fair coin n times. What is the longest streak of consecutive heads that you expect to see? We'll prove upper and lower bounds separately to show that the answer is $\Theta(\lg n)$.

We first prove that the expected length of the longest streak of heads is $O(\lg n)$. The probability that each coin flip is a head is $1/2$. Let A_{ik} be the event that a streak of heads of length at least k begins with the i th coin flip or, more precisely, the event that the k consecutive coin flips $i, i+1, \dots, i+k-1$ yield only heads, where $1 \leq k \leq n$ and $1 \leq i \leq n-k+1$. Since coin flips are mutually independent, for any given event A_{ik} , the probability that all k flips are heads is

$$\Pr\{A_{ik}\} = \frac{1}{2^k}. \quad (5.9)$$

For $k = 2 \lceil \lg n \rceil$,

$$\begin{aligned} \Pr\{A_{i, 2\lceil \lg n \rceil}\} &= \frac{1}{2^{2\lceil \lg n \rceil}} \\ &\leq \frac{1}{2^{2\lg n}} \\ &= \frac{1}{n^2}, \end{aligned}$$

and thus the probability that a streak of heads of length at least $2 \lceil \lg n \rceil$ begins in position i is quite small. There are at most $n - 2 \lceil \lg n \rceil + 1$ positions where such a streak can begin. The probability that a streak of heads of length at least $2 \lceil \lg n \rceil$ begins anywhere is therefore

$$\begin{aligned} \Pr\left\{\bigcup_{i=1}^{n-2\lceil \lg n \rceil+1} A_{i, 2\lceil \lg n \rceil}\right\} &\leq \sum_{i=1}^{n-2\lceil \lg n \rceil+1} \Pr\{A_{i, 2\lceil \lg n \rceil}\} \quad (\text{by Boole's inequality (C.21) on page 1190}) \\ &\leq \sum_{i=1}^{n-2\lceil \lg n \rceil+1} \frac{1}{n^2} \\ &< \sum_{i=1}^n \frac{1}{n^2} \\ &= \frac{1}{n}. \end{aligned} \quad (5.10)$$

We can use inequality (5.10) to bound the length of the longest streak. For $j = 0, 1, 2, \dots, n$, let L_j be the event that the longest streak of heads has length exactly j , and let L be the length of the longest streak. By the definition of expected value, we have

$$\mathbb{E}[L] = \sum_{j=0}^n j \Pr\{L_j\}. \quad (5.11)$$

We could try to evaluate this sum using upper bounds on each $\Pr\{L_j\}$ similar to those computed in inequality (5.10). Unfortunately, this method yields weak bounds. We can use some intuition gained by the above analysis to obtain a good bound, however. For no individual term in the summation in equation (5.11) are both the factors j and $\Pr\{L_j\}$ large. Why? When $j \geq 2 \lceil \lg n \rceil$, then $\Pr\{L_j\}$ is very small, and when $j < 2 \lceil \lg n \rceil$, then j is fairly small. More precisely, since the events L_j for $j = 0, 1, \dots, n$ are disjoint, the probability that a streak of heads of length at least $2 \lceil \lg n \rceil$ begins anywhere is $\sum_{j=2 \lceil \lg n \rceil}^n \Pr\{L_j\}$. Inequality (5.10) tells us that the probability that a streak of heads of length at least $2 \lceil \lg n \rceil$ begins anywhere is less than $1/n$, which means that $\sum_{j=2 \lceil \lg n \rceil}^n \Pr\{L_j\} < 1/n$. Also, noting that $\sum_{j=0}^n \Pr\{L_j\} = 1$, we have that $\sum_{j=0}^{2 \lceil \lg n \rceil - 1} \Pr\{L_j\} \leq 1$. Thus, we obtain

$$\begin{aligned}
 E[L] &= \sum_{j=0}^n j \Pr\{L_j\} \\
 &= \sum_{j=0}^{2 \lceil \lg n \rceil - 1} j \Pr\{L_j\} + \sum_{j=2 \lceil \lg n \rceil}^n j \Pr\{L_j\} \\
 &< \sum_{j=0}^{2 \lceil \lg n \rceil - 1} (2 \lceil \lg n \rceil) \Pr\{L_j\} + \sum_{j=2 \lceil \lg n \rceil}^n n \Pr\{L_j\} \\
 &= 2 \lceil \lg n \rceil \sum_{j=0}^{2 \lceil \lg n \rceil - 1} \Pr\{L_j\} + n \sum_{j=2 \lceil \lg n \rceil}^n \Pr\{L_j\} \\
 &< 2 \lceil \lg n \rceil \cdot 1 + n \cdot \frac{1}{n} \\
 &= O(\lg n).
 \end{aligned}$$

The probability that a streak of heads exceeds $r \lceil \lg n \rceil$ flips diminishes quickly with r . Let's get a rough bound on the probability that a streak of at least $r \lceil \lg n \rceil$ heads occurs, for $r \geq 1$. The probability that a streak of at least $r \lceil \lg n \rceil$ heads starts in position i is

$$\begin{aligned}
 \Pr\{A_{i, r \lceil \lg n \rceil}\} &= \frac{1}{2^{r \lceil \lg n \rceil}} \\
 &\leq \frac{1}{n^r}.
 \end{aligned}$$

A streak of at least $r \lceil \lg n \rceil$ heads cannot start in the last $n - r \lceil \lg n \rceil + 1$ flips, but let's overestimate the probability of such a streak by allowing it to start anywhere within the n coin flips. Then the probability that a streak of at least $r \lceil \lg n \rceil$ heads

occurs is at most

$$\begin{aligned} \Pr \left\{ \bigcup_{i=1}^n A_{i,r[\lg n]} \right\} &\leq \sum_{i=1}^n \Pr \{A_{i,r[\lg n]}\} && \text{(by Boole's inequality (C.21))} \\ &\leq \sum_{i=1}^n \frac{1}{n^r} \\ &= \frac{1}{n^{r-1}}. \end{aligned}$$

Equivalently, the probability is at least $1 - 1/n^{r-1}$ that the longest streak has length less than $r \lceil \lg n \rceil$.

As an example, during $n = 1000$ coin flips, the probability of encountering a streak of at least $2 \lceil \lg n \rceil = 20$ heads is at most $1/n = 1/1000$. The chance of a streak of at least $3 \lceil \lg n \rceil = 30$ heads is at most $1/n^2 = 1/1,000,000$.

Let's now prove a complementary lower bound: the expected length of the longest streak of heads in n coin flips is $\Omega(\lg n)$. To prove this bound, we look for streaks of length s by partitioning the n flips into approximately n/s groups of s flips each. If we choose $s = \lfloor (\lg n)/2 \rfloor$, we'll see that it is likely that at least one of these groups comes up all heads, which means that it's likely that the longest streak has length at least $s = \Omega(\lg n)$. We'll then show that the longest streak has expected length $\Omega(\lg n)$.

Let's partition the n coin flips into at least $\lfloor n / \lfloor (\lg n)/2 \rfloor \rfloor$ groups of $\lfloor (\lg n)/2 \rfloor$ consecutive flips and bound the probability that no group comes up all heads. By equation (5.9), the probability that the group starting in position i comes up all heads is

$$\begin{aligned} \Pr \{A_{i, \lfloor (\lg n)/2 \rfloor}\} &= \frac{1}{2^{\lfloor (\lg n)/2 \rfloor}} \\ &\geq \frac{1}{\sqrt{n}}. \end{aligned}$$

The probability that a streak of heads of length at least $\lfloor (\lg n)/2 \rfloor$ does not begin in position i is therefore at most $1 - 1/\sqrt{n}$. Since the $\lfloor n / \lfloor (\lg n)/2 \rfloor \rfloor$ groups are formed from mutually exclusive, independent coin flips, the probability that every one of these groups *fails* to be a streak of length $\lfloor (\lg n)/2 \rfloor$ is at most

$$\begin{aligned} (1 - 1/\sqrt{n})^{\lfloor n / \lfloor (\lg n)/2 \rfloor \rfloor} &\leq (1 - 1/\sqrt{n})^{n / \lfloor (\lg n)/2 \rfloor - 1} \\ &\leq (1 - 1/\sqrt{n})^{2n / \lg n - 1} \\ &\leq e^{-(2n / \lg n - 1) / \sqrt{n}} \\ &= O(e^{-\ln n}) \\ &= O(1/n). \end{aligned} \tag{5.12}$$

For this argument, we used inequality (3.14), $1 + x \leq e^x$, on page 66 and the fact, which you may verify, that $(2n/\lg n - 1)/\sqrt{n} \geq \ln n$ for sufficiently large n .

We want to bound the probability that the longest streak equals or exceeds $\lfloor (\lg n)/2 \rfloor$. To do so, let L be the event that the longest streak of heads equals or exceeds $s = \lfloor (\lg n)/2 \rfloor$. Let $\bar{L}$ be the complementary event, that the longest streak of heads is strictly less than s , so that $\Pr\{L\} + \Pr\{\bar{L}\} = 1$. Let F be the event that every group of s flips fails to be a streak of s heads. By inequality (5.12), we have $\Pr\{F\} = O(1/n)$. If the longest streak of heads is less than s , then certainly every group of s flips fails to be a streak of s heads, which means that event $\bar{L}$ implies event F . Of course, event F could occur even if event $\bar{L}$ does not (for example, if a streak of s or more heads crosses over the boundary between two groups), and so we have $\Pr\{\bar{L}\} \leq \Pr\{F\} = O(1/n)$. Since $\Pr\{L\} + \Pr\{\bar{L}\} = 1$, we have that

$$\begin{aligned} \Pr\{L\} &= 1 - \Pr\{\bar{L}\} \\ &\geq 1 - \Pr\{F\} \\ &= 1 - O(1/n) . \end{aligned}$$

That is, the probability that the longest streak equals or exceeds $\lfloor (\lg n)/2 \rfloor$ is

$$\sum_{j=\lfloor (\lg n)/2 \rfloor}^n \Pr\{L_j\} \geq 1 - O(1/n) . \quad (5.13)$$

We can now calculate a lower bound on the expected length of the longest streak, beginning with equation (5.11) and proceeding in a manner similar to our analysis of the upper bound:

$$\begin{aligned} E[L] &= \sum_{j=0}^n j \Pr\{L_j\} \\ &= \sum_{j=0}^{\lfloor (\lg n)/2 \rfloor - 1} j \Pr\{L_j\} + \sum_{j=\lfloor (\lg n)/2 \rfloor}^n j \Pr\{L_j\} \\ &\geq \sum_{j=0}^{\lfloor (\lg n)/2 \rfloor - 1} 0 \cdot \Pr\{L_j\} + \sum_{j=\lfloor (\lg n)/2 \rfloor}^n \lfloor (\lg n)/2 \rfloor \Pr\{L_j\} \\ &= 0 \cdot \sum_{j=0}^{\lfloor (\lg n)/2 \rfloor - 1} \Pr\{L_j\} + \lfloor (\lg n)/2 \rfloor \sum_{j=\lfloor (\lg n)/2 \rfloor}^n \Pr\{L_j\} \\ &\geq 0 + \lfloor (\lg n)/2 \rfloor (1 - O(1/n)) \quad (\text{by inequality (5.13)}) \\ &= \Omega(\lg n) . \end{aligned}$$